Chapter 6

Green's identities and Poisson problems

Green's identities - first identity

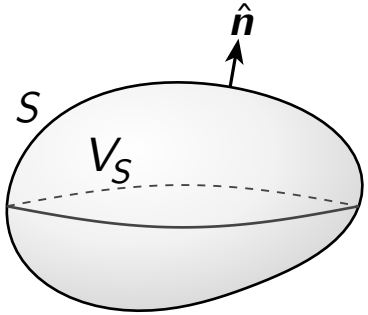
Consider two scalar functions $\varphi, \psi \in C^2$ in a domain V_S bounded by the closed surface S

C^2: continuous, with continuous first and second derivatives

Divergence of the vector field $\boldsymbol{F} = \varphi \nabla \psi$:

$$\nabla \cdot (\varphi \nabla \psi) = \nabla \varphi \cdot \nabla \psi + \varphi \nabla^2 \psi$$

vector identity: $\nabla \cdot (f \nabla g) = \nabla f \cdot \nabla g + f \nabla^2 g$



Integrating over V_S:

$$\int_{V_S} \nabla \cdot (\varphi \nabla \psi) \, \mathrm{d}V = \int_{V_S} (\nabla \varphi \cdot \nabla \psi + \varphi \nabla^2 \psi) \, \mathrm{d}V$$

Green's identities - first identity (cont.)

From the volume integral to a surface integral:

$$\int_{V_S} \nabla \cdot (\varphi \nabla \psi) \, \mathrm{d}V = \int_{V_S} (\nabla \varphi \cdot \nabla \psi + \varphi \nabla^2 \psi) \, \mathrm{d}V$$

$$\downarrow \text{ divergence theorem}$$

$$\oint_S \varphi \nabla \psi \cdot \hat{\boldsymbol{n}} \, \mathrm{d}S = \int_{V_S} (\nabla \varphi \cdot \nabla \psi + \varphi \nabla^2 \psi) \, \mathrm{d}V$$

$$\downarrow \nabla \psi \cdot \hat{\boldsymbol{n}} = \partial \psi / \partial n \text{ (def. of gradient: derivative along } \hat{\boldsymbol{n}})$$

$$\oint_S \varphi \frac{\partial \psi}{\partial n} \, \mathrm{d}S = \int_{V_S} \nabla \varphi \cdot \nabla \psi \, \mathrm{d}V + \int_{V_S} \varphi \nabla^2 \psi \, \mathrm{d}V$$

Green's identities - integration by parts

Green's first identity:

$$\oint_S \varphi \frac{\partial \psi}{\partial n} \, \mathrm{d}S = \int_{V_S} \nabla \varphi \cdot \nabla \psi \, \mathrm{d}V + \int_{V_S} \varphi \nabla^2 \psi \, \mathrm{d}V$$

Reordering, and comparing with 1D:

$$\int_{V_S} \nabla \varphi \cdot \nabla \psi \, \mathrm{d}V = \oint_S \varphi \frac{\partial \psi}{\partial n} \, \mathrm{d}S - \int_{V_S} \varphi \nabla^2 \psi \, \mathrm{d}V \quad (1)$$

$$\int_a^b f' g \, \mathrm{d}x = [f g]_a^b - \int_a^b f g' \, \mathrm{d}x$$

(1) is the **multi-dimensional** integration by parts: $[f g]_a^b$ becomes the surface integral on S

Green's identities - second identity

Recalling (1):

$$\int_{V_S} \nabla\varphi \cdot \nabla\psi \, \mathrm{d}V = \oint_S \varphi \frac{\partial\psi}{\partial n} \, \mathrm{d}S - \int_{V_S} \varphi \nabla^2\psi \, \mathrm{d}V \qquad (1)$$

Same steps with $\boldsymbol{F} = \psi\nabla\varphi$ (φ and ψ swapped):

$$\int_{V_S} \nabla\psi \cdot \nabla\varphi \, \mathrm{d}V = \oint_S \psi \frac{\partial\varphi}{\partial n} \, \mathrm{d}S - \int_{V_S} \psi \nabla^2\varphi \, \mathrm{d}V \qquad (2)$$

Green's identities - second identity (cont.)

Subtracting (1) − (2) member by member: the left-hand sides are equal

$$\cancel{\int_{V_s} (\nabla\varphi \cdot \nabla\psi - \nabla\psi \cdot \nabla\varphi)\,\mathrm{d}V} = \oint_S \left(\varphi \frac{\partial\psi}{\partial n} - \psi \frac{\partial\varphi}{\partial n}\right)\mathrm{d}S - \int_{V_s} (\varphi\nabla^2\psi - \psi\nabla^2\varphi)\,\mathrm{d}V$$

The left-hand side is zero, so:

$$\oint_S \left(\varphi \frac{\partial\psi}{\partial n} - \psi \frac{\partial\varphi}{\partial n}\right)\mathrm{d}S = \int_{V_s} (\varphi\nabla^2\psi - \psi\nabla^2\varphi)\,\mathrm{d}V$$

Green's second identity: converts volume integrals of Laplacians into surface integrals

Harmonic functions - definition

Consider a domain V_S and a function $\varphi \in C^2$

> φ is **harmonic** in V_S if $\nabla^2 \varphi = \nabla \cdot (\nabla \varphi) = 0 \quad \forall \boldsymbol{x} \in V_S$

Laplacian: divergence of the gradient

Examples

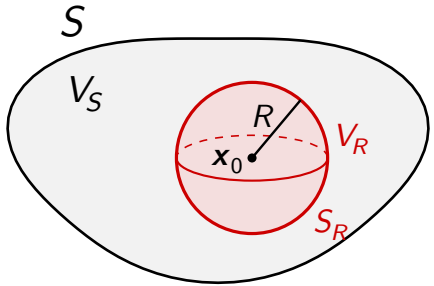
- constant functions: $\varphi = k$, $k \in \mathbb{R}$
- first-order polynomials: $\varphi = ax + by + cz + d$
- $\varphi = 1/r$, with $r = |\boldsymbol{x} - \boldsymbol{x}_0|$, for $\boldsymbol{x} \neq \boldsymbol{x}_0$

Physical example the electric potential in a region **without charge**: $\nabla^2 \varphi = -\rho/\varepsilon = 0$

Mean value theorem for harmonic functions

φ harmonic in V_S. Consider a sphere V_R of radius R, centred at $\boldsymbol{x}_0$, bounded by S_R and contained in V_S

$$\varphi(\boldsymbol{x}_0) = \underbrace{\frac{1}{4\pi R^2} \oint_{S_R} \varphi \, \mathrm{d}S}_{\text{average on } S_R} = \underbrace{\frac{1}{\frac{4}{3}\pi R^3} \int_{V_R} \varphi \, \mathrm{d}V}_{\text{average on } V_R}$$



The value at the centre is the average of the surrounding values, on every sphere $V_R \subset V_S$

proof: from Green's identities with $\psi = 1/r$ (lecture notes)

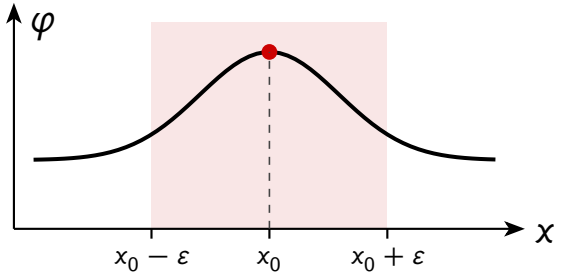
Harmonic functions - no internal extrema

Corollary K_1 A harmonic function has no local extrema at the internal points of its domain

By contradiction: assume that there **is** a local maximum at an internal point $\boldsymbol{x}_0$

Local maximum in $\boldsymbol{x}_0$: $\exists$ a small sphere V_ε around $\boldsymbol{x}_0$ in which φ is smaller than $\varphi(\boldsymbol{x}_0)$ everywhere except in $\boldsymbol{x}_0$

$$\varphi(\boldsymbol{x}) < \varphi(\boldsymbol{x}_0) \quad \forall \boldsymbol{x} \in V_\varepsilon, \boldsymbol{x} \neq \boldsymbol{x}_0$$



right: a 1D sketch, V_ε becomes $[x_0 - \varepsilon, x_0 + \varepsilon]$

Harmonic functions - no internal extrema (proof)

Assumed: $\varphi(\boldsymbol{x}) < \varphi(\boldsymbol{x}_0)$ for all the other points $\boldsymbol{x}$ of the sphere V_ε

Define the auxiliary function

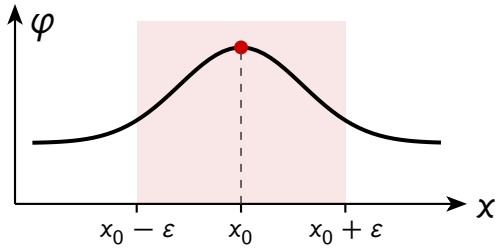
$$\tilde{\varphi}(\boldsymbol{x}) = \varphi(\boldsymbol{x}_0) - \varphi(\boldsymbol{x})$$

From the assumption:

$$\tilde{\varphi}(\boldsymbol{x}) > 0 \quad \forall \boldsymbol{x} \neq \boldsymbol{x}_0, \qquad \tilde{\varphi}(\boldsymbol{x}_0) = 0$$

$\tilde{\varphi}$ is harmonic (linearity of ∇^2):

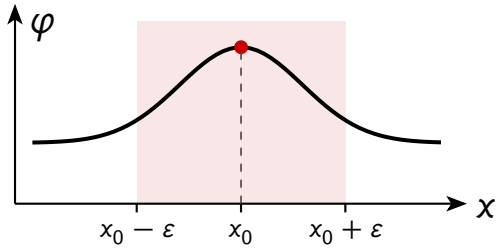
$$\nabla^2 \tilde{\varphi} = \underbrace{\nabla^2 \varphi(\boldsymbol{x}_0)}_{\text{constant: } 0} - \underbrace{\nabla^2 \varphi}_{\text{harmonic: } 0} = 0$$



Harmonic functions - no internal extrema (proof, cont.)

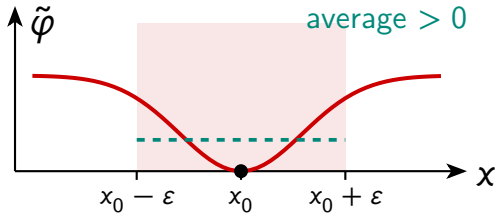
$\tilde{\varphi}$ harmonic: mean value theorem on V_ε

$$\underbrace{\tilde{\varphi}(\boldsymbol{x}_0)}_{=0 \text{ by definition}} = \underbrace{\frac{1}{\frac{4}{3}\pi\varepsilon^3}\int_{V_\varepsilon}\tilde{\varphi}(\boldsymbol{x})\,\mathrm{d}V}_{>0 \,:\, \text{average of positive values}}$$



$0 > 0$: impossible. The assumption is false:

> no local maximum inside V_S

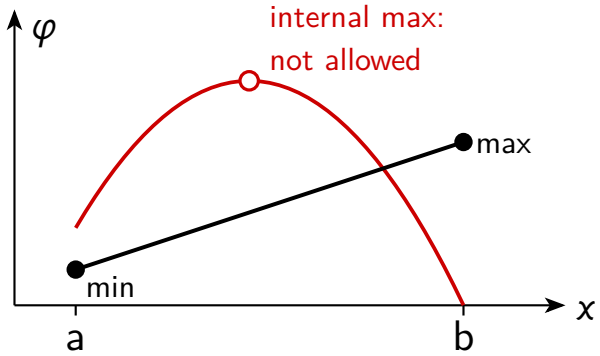


for a minimum: same argument with $\tilde{\varphi} = \varphi(\boldsymbol{x}) - \varphi(\boldsymbol{x}_0)$

Harmonic functions - extrema on the boundary

Corollary K_2 The extrema of a harmonic function in V_S lie on the **boundary** S

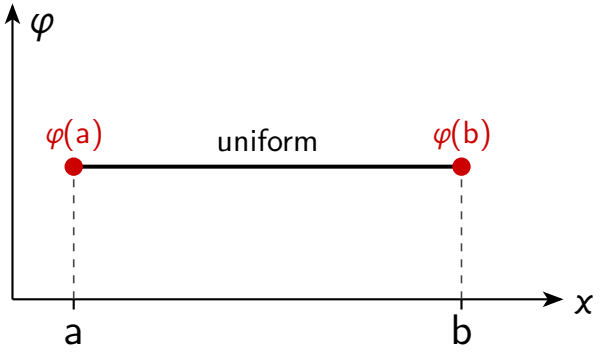
Example in 1D, $\varphi'' = 0$ in $[a, b] \Rightarrow \varphi$ is a straight line: min and max at the ends a and b.
An internal maximum (red) is excluded by K_1



Harmonic functions - uniform boundary values

Corollary K_3 If a harmonic function is **uniform** on the whole boundary S, it is uniform in V_S

(by K_2: max and min lie on S, where φ has a single value)



Example in 1D, the boundary is $\{a, b\}$: $\varphi(a) = \varphi(b) \Rightarrow$ the straight line is horizontal, φ uniform

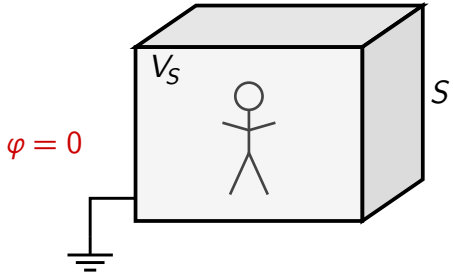
Example the Faraday cage (next slide)

Harmonic functions - example: Faraday cage

Example a closed conducting box connected to ground: $\varphi = 0$ on S. No charge inside

$\rho = 0$ in V_S:

$$\nabla^2 \varphi = -\frac{\rho}{\varepsilon_0} = 0 \quad \Rightarrow \quad \varphi \text{ harmonic in } V_S$$



$\varphi = 0$ on the whole $S \Rightarrow$ by K_3, $\varphi = 0$ in V_S:

$$\boldsymbol{E} = -\nabla \varphi = 0$$

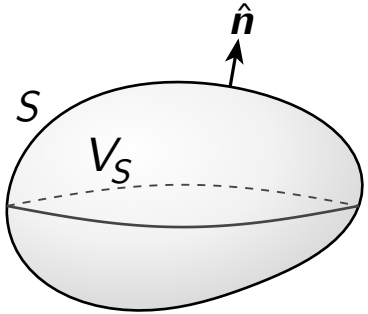
Box at any other uniform potential, e.g. $\varphi = 1$ kV on S: by K_3, $\varphi = 1$ kV in V_S, and again $\boldsymbol{E} = -\nabla \varphi = 0$: **no electric field inside**

Poisson problems - uniqueness with Dirichlet conditions

Setting: uniqueness already proven for Maxwell's equations; now for problems written for a **potential** φ

Poisson equation (t: known source, e.g. $t = -\rho/\varepsilon$), with φ assigned on the whole boundary (**Dirichlet** condition):

$$\begin{cases} \nabla^2 \varphi = t & \forall \boldsymbol{x} \in V_S \\ \varphi = \varphi_0 & \forall \boldsymbol{x} \in S \end{cases} \qquad (1)$$



By contradiction: suppose two different solutions φ_1, φ_2 of (1)

Goal: show that their difference is zero

$$\varphi_3 = \varphi_1 - \varphi_2 = 0$$

Poisson problems - uniqueness with Dirichlet conditions (cont.)

Applying (1) to φ_1 and to φ_2:

$$\begin{cases} \nabla^2 \varphi_1 = t & \text{in } V_S \\ \varphi_1 = \varphi_0 & \text{on } S \end{cases} \quad (2) \qquad \begin{cases} \nabla^2 \varphi_2 = t & \text{in } V_S \\ \varphi_2 = \varphi_0 & \text{on } S \end{cases} \quad (3)$$

Subtracting (2) $-$ (3) member by member (linearity of ∇^2):

$$\begin{aligned} \text{in } V_S: \quad & \nabla^2 \varphi_1 - \nabla^2 \varphi_2 = t - t = 0 \quad \Rightarrow \quad \nabla^2 \varphi_3 = 0 \\ \text{on } S: \quad & \varphi_1 - \varphi_2 = \varphi_0 - \varphi_0 = 0 \quad \Rightarrow \quad \varphi_3 = 0 \end{aligned}$$

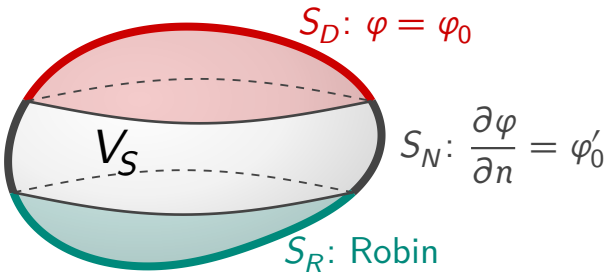
φ_3 harmonic, and uniformly zero on S: by K_3, $\varphi_3 = 0$ in V_S

$\varphi_3 = \varphi_1 - \varphi_2 = 0 \quad \Rightarrow \quad \varphi_1 = \varphi_2$: the solution is **unique**

Poisson problems - Dirichlet, Neumann and Robin conditions

Besides Dirichlet, two other very useful kinds of boundary condition

- **Dirichlet** on S_D: $\varphi = \varphi_0$
- **Neumann** on S_N: $\partial\varphi/\partial n = \varphi_0'$
  electric potential: $E_n = -\nabla\varphi \cdot \hat{\boldsymbol{n}} = -\partial\varphi/\partial n$, i.e. the normal component of $\boldsymbol{E}$ is assigned
- **Robin** on S_R: a combination of φ and $\partial\varphi/\partial n$
  e.g. the connection with an external circuit

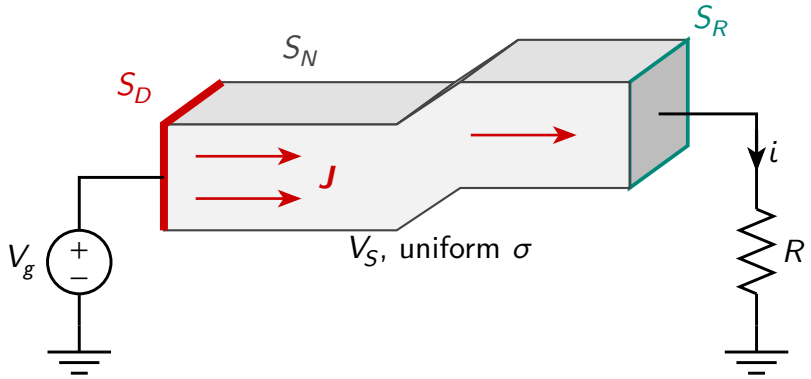


Where φ is not assigned, φ_3 is not known to be uniform on the whole S: K_3 **cannot** be used

Next: an example where all three conditions appear, then one uniqueness proof for all of them

Steady-state conduction - the problem

Setting: a conductor V_S of irregular shape, with uniform and known σ [S/m], in DC, connected to an external circuit. **Goal**: a formulation for $\boldsymbol{J}$ and $\boldsymbol{E}$ inside the conductor



Regular shape (e.g. a uniform cylinder): analytical solution. Irregular shape: **numerical** analysis, which needs a formulation first

Steady-state conduction - field equations

Steady state, $\partial/\partial t = 0$, at the internal points of V_S:

$$\nabla \times \boldsymbol{E} = -\cancel{\frac{\partial \boldsymbol{B}}{\partial t}} = 0, \qquad \nabla \cdot \left(\boldsymbol{J} + \cancel{\frac{\partial \boldsymbol{D}}{\partial t}} \right) = 0, \qquad \boldsymbol{J} = \sigma \boldsymbol{E}$$

What is left:

$$\nabla \times \boldsymbol{E} = 0, \qquad \nabla \cdot \boldsymbol{J} = 0, \qquad \boldsymbol{J} = \sigma \boldsymbol{E}$$

V_S simply connected and $\nabla \times \boldsymbol{E} = 0 \Rightarrow \boldsymbol{E}$ is conservative:

$$\boldsymbol{E} = -\nabla \varphi$$

Steady-state conduction - Laplace equation

Rewriting $\nabla \cdot \boldsymbol{J} = 0$ in terms of φ:

$$\nabla \cdot \boldsymbol{J} = 0$$

$$\downarrow \quad \text{constitutive law: } \boldsymbol{J} = \sigma \boldsymbol{E}$$

$$\nabla \cdot (\sigma \boldsymbol{E}) = 0$$

$$\downarrow \quad \boldsymbol{E} = -\nabla \varphi$$

$$\nabla \cdot (-\sigma \nabla \varphi) = 0$$

$$\downarrow \quad \sigma \text{ uniform: out of the divergence}$$

$$-\sigma \nabla \cdot \nabla \varphi = 0$$

$$\downarrow \quad \text{dividing by } -\sigma;\ \nabla \cdot \nabla = \nabla^2$$

$$\nabla^2 \varphi = 0 \quad \text{Laplace equation: } \varphi \text{ harmonic in } V_S$$

Steady-state conduction - role of σ

The distribution of φ is determined **only** by the boundary conditions

With uniform σ, the conductivity plays **no role** in it

> Same in electrostatics: charge-free region, uniform ε
> $\nabla \cdot (\varepsilon \nabla \varphi) = 0 \Rightarrow \nabla^2 \varphi = 0$: ε_r has no influence on φ

Recipe at the internal points:

$$\text{solve } \nabla^2 \varphi = 0 \quad \rightarrow \quad \boldsymbol{E} = -\nabla \varphi \quad \rightarrow \quad \boldsymbol{J} = \sigma \boldsymbol{E}$$

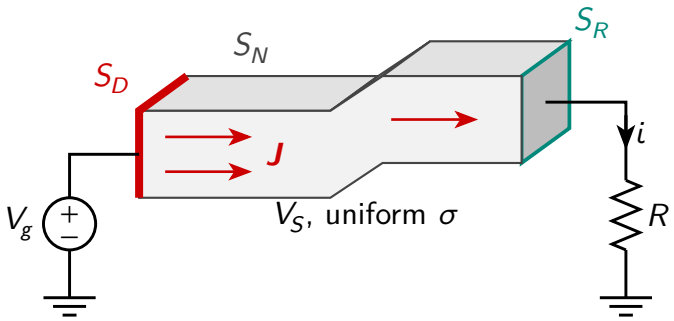
Still missing: how to set the **boundary conditions** (next slides)

Steady-state conduction - boundary conditions

S_D (electrode on the generator): $\varphi = V_g$

S_N (lateral surface): no current through it

$$\boldsymbol{J} \cdot \hat{\boldsymbol{n}} = \sigma \boldsymbol{E} \cdot \hat{\boldsymbol{n}} = -\sigma \nabla \varphi \cdot \hat{\boldsymbol{n}} = 0 \quad \overset{\div(-\sigma)}{\Rightarrow} \quad \frac{\partial \varphi}{\partial n} = 0$$



S_R (electrode on the resistor): $\varphi = V_R$

V_R is **not** known: it depends on the current i, i.e. on the solution

Steady-state conduction - condition on S_R

Express V_R through φ. Assume no skin effect: $\boldsymbol{J}$, $\boldsymbol{E}$ and $\partial\varphi/\partial n$ uniform on S_R, of area A_R

Ohm's law for the resistor, and the current through S_R:

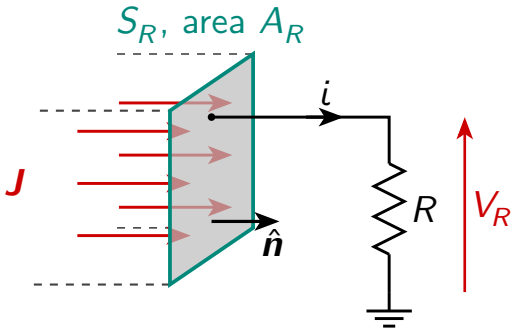
$$\varphi = V_R = Ri$$

$$\downarrow \quad i = \int_{S_R} \boldsymbol{J} \cdot \hat{\boldsymbol{n}} \, \mathrm{d}S$$

$$\varphi = R \int_{S_R} \boldsymbol{J} \cdot \hat{\boldsymbol{n}} \, \mathrm{d}S$$

$$\downarrow \quad \boldsymbol{J} = \sigma \boldsymbol{E} = -\sigma \nabla \varphi$$

$$\varphi = R \int_{S_R} (-\sigma \nabla \varphi) \cdot \hat{\boldsymbol{n}} \, \mathrm{d}S$$



Steady-state conduction - condition on S_R (cont.)

Taking $-\sigma$ (uniform) out of the integral:

$$\varphi = -R\sigma \int_{S_R} \nabla\varphi \cdot \hat{\boldsymbol{n}} \, \mathrm{d}S$$

$$\downarrow \quad \nabla\varphi \cdot \hat{\boldsymbol{n}} = \partial\varphi/\partial n \text{ (def. of gradient)}$$

$$\varphi = -R\sigma \int_{S_R} \frac{\partial\varphi}{\partial n} \, \mathrm{d}S$$

$$\downarrow \quad \partial\varphi/\partial n \text{ uniform on } S_R\text{: out of the integral}$$

$$\varphi = -R\sigma \frac{\partial\varphi}{\partial n} \int_{S_R} \mathrm{d}S = -R\sigma A_R \frac{\partial\varphi}{\partial n}$$

Steady-state conduction - Robin condition

We found, on S_R:

$$\varphi = -R\sigma A_R \frac{\partial \varphi}{\partial n}$$

Rearranging and generalizing:

$$\underbrace{1}_{\alpha} \cdot \varphi + \underbrace{R\sigma A_R}_{\beta} \frac{\partial \varphi}{\partial n} + \underbrace{0}_{\gamma} = 0$$

$$\alpha\varphi + \beta\frac{\partial \varphi}{\partial n} + \gamma = 0 \quad \textbf{Robin} \text{ boundary condition}$$

a linear combination of a Dirichlet and a Neumann condition; α, β, γ known

Steady-state conduction - formulation

Formulation of the conduction problem:

$$\begin{cases} \nabla^2 \varphi = 0 & \text{in } V_S \\ \varphi = V_g & \text{on } S_D \\ \dfrac{\partial \varphi}{\partial n} = 0 & \text{on } S_N \\ \varphi + R\sigma A_R \dfrac{\partial \varphi}{\partial n} = 0 & \text{on } S_R \end{cases}$$

V_g, R, σ, A_R: known electrical and geometrical data

Dirichlet, Neumann and Robin conditions on the three parts of S: is the solution unique?

Poisson problems - uniqueness: two solutions

By contradiction: two solutions φ_1, φ_2 of the conduction problem, with a generic source t and $\beta = R\sigma A_R$. Goal: show that $\varphi_3 = \varphi_1 - \varphi_2 = 0$

Applying the formulation to φ_1 and to φ_2:

$$\begin{cases} \nabla^2 \varphi_1 = t & \text{in } V_S \\ \varphi_1 = V_g & \text{on } S_D \\ \dfrac{\partial \varphi_1}{\partial n} = 0 & \text{on } S_N \\ \varphi_1 + \beta \dfrac{\partial \varphi_1}{\partial n} = 0 & \text{on } S_R \end{cases} \qquad \begin{cases} \nabla^2 \varphi_2 = t & \text{in } V_S \\ \varphi_2 = V_g & \text{on } S_D \\ \dfrac{\partial \varphi_2}{\partial n} = 0 & \text{on } S_N \\ \varphi_2 + \beta \dfrac{\partial \varphi_2}{\partial n} = 0 & \text{on } S_R \end{cases}$$

Poisson problems - uniqueness: difference of two solutions

Subtracting member by member (linearity), on each part of the domain:

$$
\begin{aligned}
\text{in } V_S: &\quad \nabla^2\varphi_1 - \nabla^2\varphi_2 = t - t = 0 && \Rightarrow \quad \nabla^2\varphi_3 = 0 \\
\text{on } S_D: &\quad \varphi_1 - \varphi_2 = V_g - V_g = 0 && \Rightarrow \quad \varphi_3 = 0 \\
\text{on } S_N: &\quad \frac{\partial\varphi_1}{\partial n} - \frac{\partial\varphi_2}{\partial n} = 0 - 0 = 0 && \Rightarrow \quad \frac{\partial\varphi_3}{\partial n} = 0 \\
\text{on } S_R: &\quad (\varphi_1 - \varphi_2) + \beta\left(\frac{\partial\varphi_1}{\partial n} - \frac{\partial\varphi_2}{\partial n}\right) = 0 \Rightarrow && \varphi_3 + \beta\frac{\partial\varphi_3}{\partial n} = 0
\end{aligned}
$$

The known data (V_g, the source t) cancel: φ_3 solves the problem with zero data

K_3 cannot be used: apply Green's first identity with $\varphi = \psi = \varphi_3$

Poisson problems - uniqueness: Green's first identity

Green's first identity with $\varphi = \psi = \varphi_3$, harmonic:

$$\oint_S \varphi_3 \frac{\partial \varphi_3}{\partial n} \, \mathrm{d}S = \int_{V_S} |\nabla \varphi_3|^2 \, \mathrm{d}V + \int_{V_S} \varphi_3 \cancel{\nabla^2 \varphi_3} \, \mathrm{d}V$$

Split $S = S_D \cup S_N \cup S_R$ and look at the integrand $\varphi_3 \, \partial\varphi_3/\partial n$ on each part:

part	condition on φ_3	$\varphi_3 \, \partial\varphi_3/\partial n$
S_D	$\varphi_3 = 0$	0
S_N	$\partial\varphi_3/\partial n = 0$	0
S_R	$\partial\varphi_3/\partial n = -\varphi_3/\beta$	$-\varphi_3^2/\beta \leq 0$

Substituting on each part: the integrals on S_D and S_N vanish, only S_R is left

$$\underbrace{-\frac{1}{\beta} \int_{S_R} \varphi_3^2 \, \mathrm{d}S}_{\leq 0} = \underbrace{\int_{V_S} |\nabla \varphi_3|^2 \, \mathrm{d}V}_{\geq 0}$$

Poisson problems - uniqueness: conclusion

A quantity ≤ 0 equals a quantity ≥ 0: both are zero

$$\int_{V_S} |\nabla \varphi_3|^2 \, \mathrm{d}V = 0 \quad \Rightarrow \quad \nabla \varphi_3 = 0 \text{ in } V_S \quad (\varphi_3 \text{ uniform})$$

φ_3 uniform and $\varphi_3 = 0$ on $S_D \Rightarrow \varphi_3 = 0$ everywhere

> $\varphi_1 = \varphi_2$: the conduction problem has a **unique** solution

The same proof applies to any subset of the three conditions:

- Dirichlet only, or Dirichlet + Neumann: **unique**
- Neumann on the whole S (no Dirichlet or Robin part): φ_3 uniform, but no condition fixes its value: unique **up to a constant** (for the potential: the reference is free, $\boldsymbol{E} = -\nabla\varphi$ is unique)

Poisson problems - general Robin condition

General Robin condition on S_R: $\alpha\varphi + \beta\partial\varphi/\partial n + \gamma = 0$

For the difference: $\alpha\varphi_3 + \beta\partial\varphi_3/\partial n = 0$ (γ cancels), so on S_R

$$\varphi_3 \frac{\partial \varphi_3}{\partial n} = -\frac{\alpha}{\beta}\varphi_3^2 \quad \Rightarrow \quad -\frac{\alpha}{\beta}\int_{S_R} \varphi_3^2 \, \mathrm{d}S = \int_{V_S} |\nabla \varphi_3|^2 \, \mathrm{d}V$$

$\alpha/\beta > 0$: left-hand side ≤ 0, the proof works as before

$\alpha/\beta < 0$: both sides ≥ 0, a $\varphi_3 \neq 0$ can satisfy the equality: uniqueness is **not** guaranteed

> With a Robin condition the solution is unique for $\alpha/\beta > 0$
> (conduction: $\alpha = 1$, $\beta = R\sigma A_R > 0$)

Example convective heat exchange at a wall: $-k\ \partial T/\partial n = h(T - T_\infty)$, i.e. $hT + k\ \partial T/\partial n - hT_\infty = 0$: $\alpha = h$, $\beta = k$, $\alpha/\beta > 0$